Practical No.: 4

Title: Creating a Student Registration Form in ReactJS

Aim:

To develop a **student registration form** in ReactJS using form components like **input fields, radio buttons, select dropdowns, and a submit button**.

Theory

A **registration form** is used to collect user details. In **ReactJS**, form handling is done using **controlled components**, where form inputs are managed using **state**. We use the `useState` hook to track user inputs and update the UI dynamically.

Concepts Covered:

- Controlled Components
- Handling input fields (text, email, number)
- Radio buttons and dropdown selection
- Handling form submission

Syntax for Handling Input in ReactJS:

```
const [value, setValue] = React.useState("");

const handleChange = (event) => {
    setValue(event.target.value);
};
```

.ReactJS Code

Step 1: Create `index.html`

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Student Registration Form</title>
    <script src="https://unpkg.com/react@18/umd/react.development.js"></script>
    <script src="https://unpkg.com/react-dom@18/umd/react-
dom.development.js"></script>
    <script src="https://unpkg.com/babel-standalone@6/babel.min.js"></script>
</head>
<body>
    <h2>Practical 4: Student Registration Form</h2>
    <div id="root"></div>

    <script type="text/babel" src="app.js"></script>
</body>
</html>
```

Step 2: Create a JavaScript file (app.js)

```
function StudentRegistrationForm() {
    const [name, setName] = React.useState("");
    const [email, setEmail] = React.useState("");
    const [age, setAge] = React.useState("");
    const [gender, setGender] = React.useState("");
    const [course, setCourse] = React.useState("BCA");

    const handleSubmit = (event) => {
        event.preventDefault();
        alert(`Student Registered!\n\nName: ${name}\nEmail: ${email}\nAge:
${age}\nGender: ${gender}\nCourse: ${course}`);
    };

    return (
        <div>
            <h3>Student Registration Form</h3>
            <form onSubmit={handleSubmit}>
                <label>
                    Full Name:
                    <input
                        type="text"
                        value={name}
                        onChange={(e) => setName(e.target.value)}
                        required
                    />
                </label>
                <br /><br />

                <label>
                    Email:
                    <input
                        type="email"
                        value={email}
                        onChange={(e) => setEmail(e.target.value)}
                        required
                    />
                </label>
                <br /><br />

                <label>
                    Age:
                    <input
                        type="number"
                        value={age}
                        onChange={(e) => setAge(e.target.value)}
                        required
                    />
                </label>
                <br /><br />

                <label>
```

```
                    Gender:
                    <input
                        type="radio"
                        name="gender"
                        value="Male"
                        onChange={(e) => setGender(e.target.value)}
                        required
                    /> Male
                    <input
                        type="radio"
                        name="gender"
                        value="Female"
                        onChange={(e) => setGender(e.target.value)}
                        required
                    /> Female
                </label>
                <br /><br />

                <label>
                    Select Course:
                    <select value={course} onChange={(e) =>
setCourse(e.target.value)}>
                        <option value="BCA">BCA</option>
                        <option value="MCA">MCA</option>
                        <option value="BSc IT">BSc IT</option>
                        <option value="MSc IT">MSc IT</option>
                    </select>
                </label>
                <br /><br />

                <button type="submit">Register</button>
            </form>
        </div>
    );
}

const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(<StudentRegistrationForm />);
```